



EEE-301

Data and Telecommunication

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Switching

- Switching, as applied to networking and IT, is the practice of directing a signal or data element toward a particular hardware destination.
- Switching may be applied in various formats and can function in diverse ways within a greater network infrastructure.
- **Network switching** is the process of channeling data received from any number of input ports, to another designated port that will transmit the data to its desired destination.

Example of Switching

- **For example:** whenever a telephone call is placed, there are numerous junctions in the communication path that perform this movement of data from one network onto another network. One of another example is gateway, that can be used by Internet Service Providers (ISP) to deliver a signal to another Internet Service Providers (ISP). For exchange of information between different locations various types of Switching Techniques are used in Networking.

Types of Switching

- There are basically three types of switching methods are available.

1. Circuit Switching.

2. Packet Switching.

3. Message Switching.

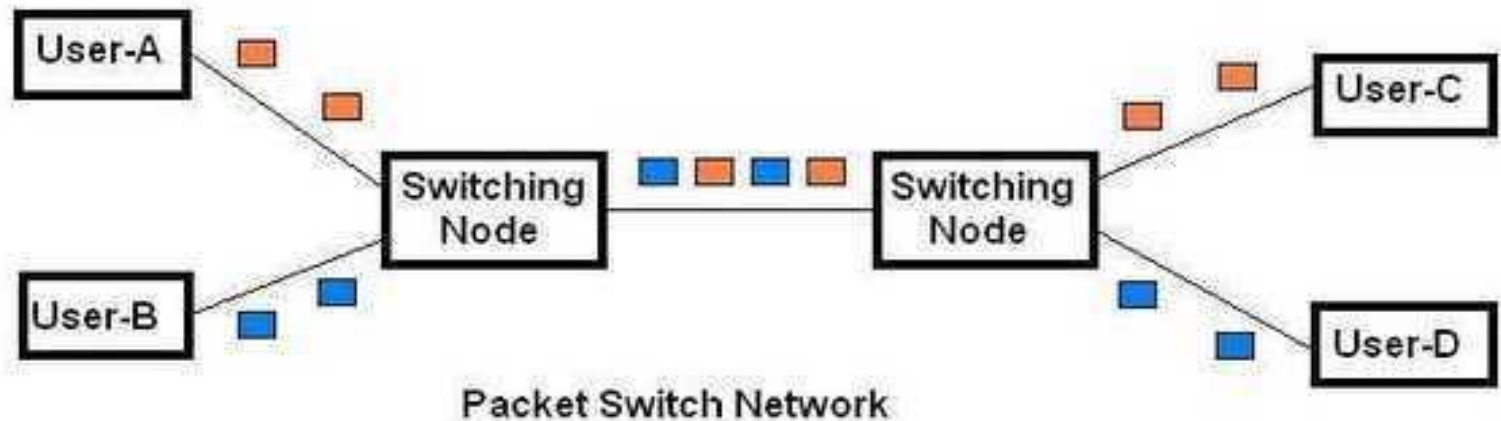
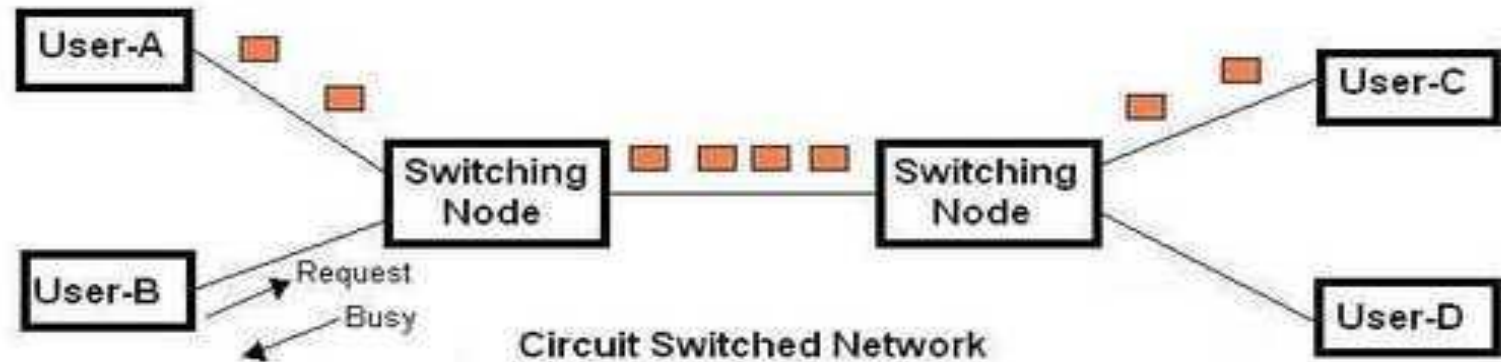
1. Circuit Switching

- **Circuit switching** is a method of implementing a telecommunications network in which two network nodes establish a dedicated communications channel (circuit) through the network before the nodes may communicate.
- The circuit guarantees the full bandwidth of the channel and remains connected for the duration of the communication session.
- The circuit functions as if the nodes were physically connected as with an electrical circuit.

2. Packet Switching

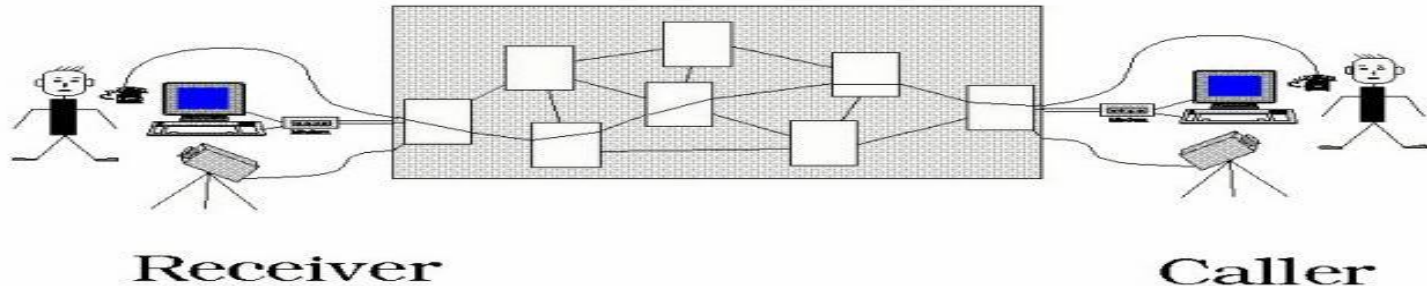
Packet-switched networks move data in separate, small blocks (packets) based on the destination address in each packet. When received, packets are reassembled in the proper sequence to make up the message.

Circuit Switching VS Packet Switching

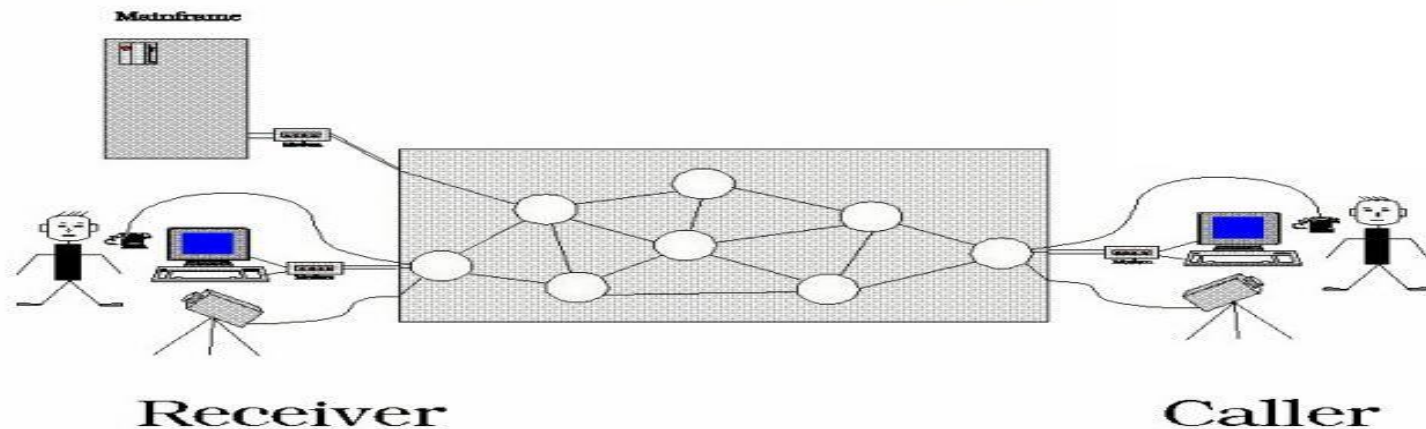


Circuit Switching VS Packet Switching

Circuit Switched Network

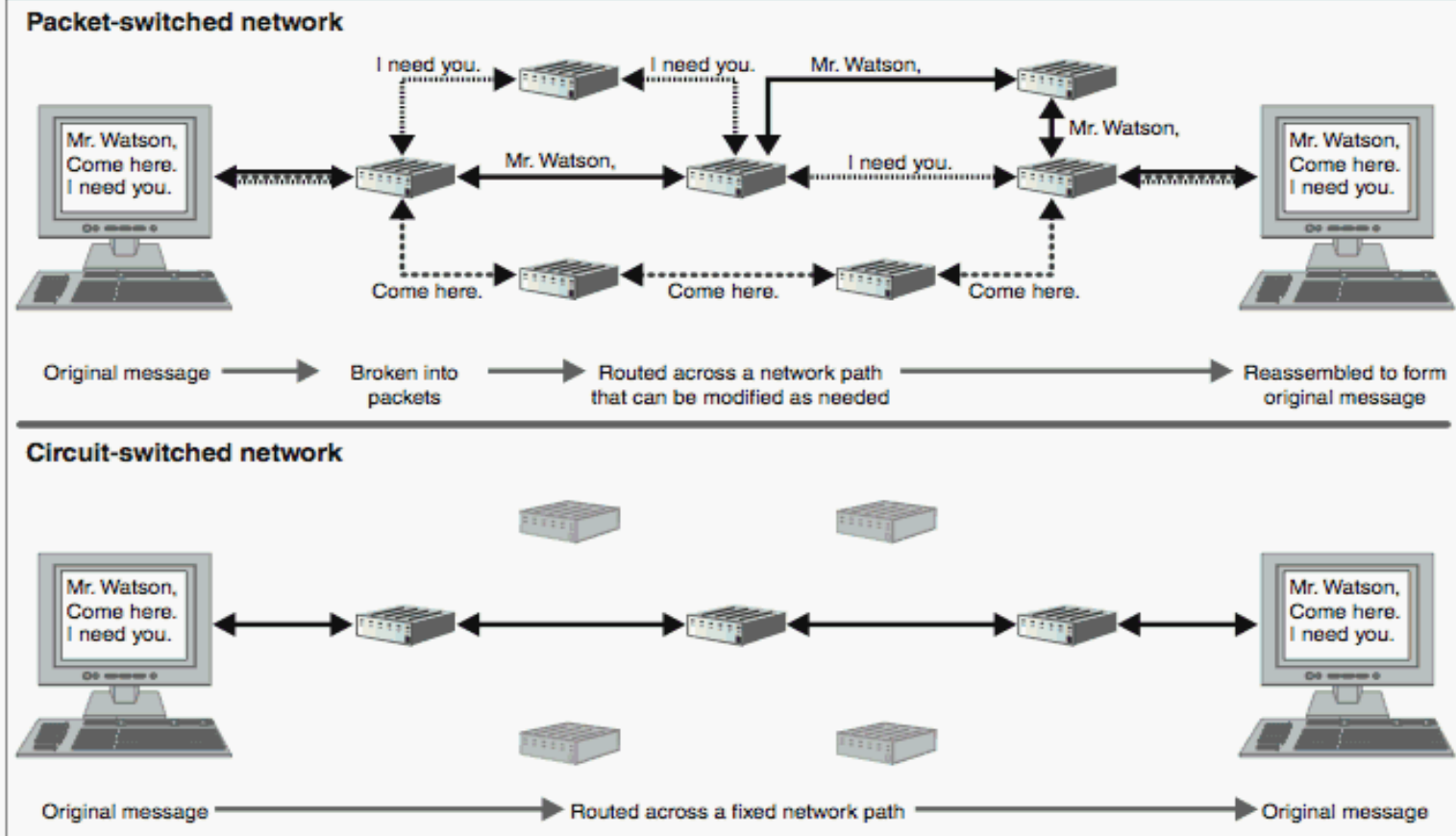


Packet Switched Network



Circuit Switching VS Packet Switching

Figure 1: Packet Switching vs. Circuit Switching



Source: GAO analysis; Art Explosion (images).

Circuit Switching VS Packet Switching(1)

- **Key Differences Between Circuit Switching and Packet Switching**
- Circuit Switching is connection oriented that means a path is established between source and destination before the transmission occurs. On the other hand, Packet Switching is Connectionless that means a dynamic route is decided for each packet while transmission.
- Circuit Switching was originally designed for voice communication whereas, Packet Switching was originally designed for data communication.
- Circuit Switching is inflexible as once a path is established for transmission, it doesn't change while the duration of the session. On the other hand, Packet Switching is flexible as each packet may travel through a different route to reach its destination.

Circuit Switching VS Packet Switching (2)

- **Key Differences Between Circuit Switching and Packet Switching**
- In packet switching, as each packet travels a different path hence, the packet are received out of order at the receiver side and later arranged in order. On the other hand, in circuit switching the entire message is received as it is as sent from a sender to receiver.
- Space Division Switching or Time-Division Switching can be used to implement Circuit Switching whereas, Packet Switching can be implemented using two approaches Datagram Approach and Virtual Circuit Approach.
- Circuit Switching is always implemented at physical layer whereas, Packet Switching is implemented on the network layer.

3. Message Switching

- Message switching does not set up a dedicated channel (or circuit) between the sender and recipient during the communication session.
- In Message Switching each message is treated as an independent blocks.
- In this type of networking, each message is then transmitted from first network device to second network device through the internetwork i.e. message is transmitted from the sender to intermediary device.
- The intermediate device stores the message for a time being, after inspects it for errors, intermediate device transmitting the message to the next node with its routing information.
- Because of this reason message switching networks are called store and forward networks in networking.

3. Message Switching

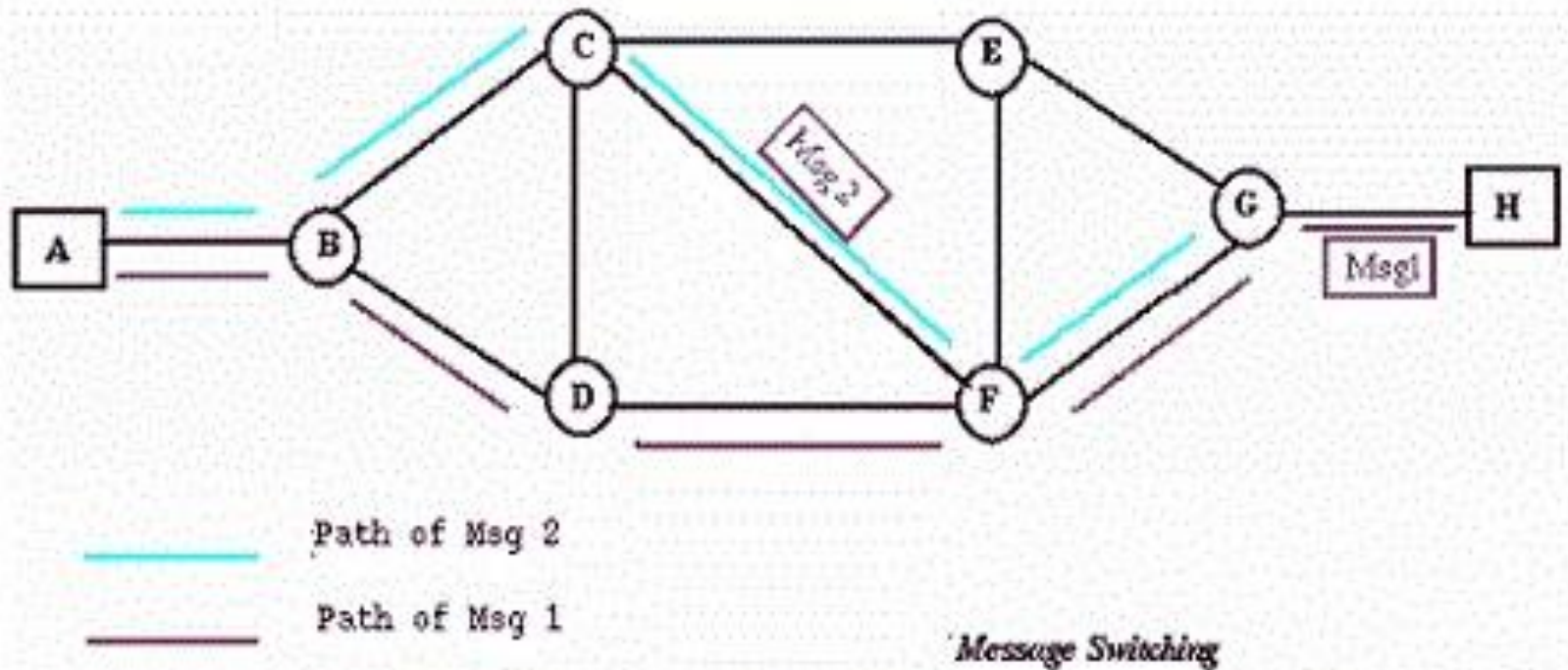
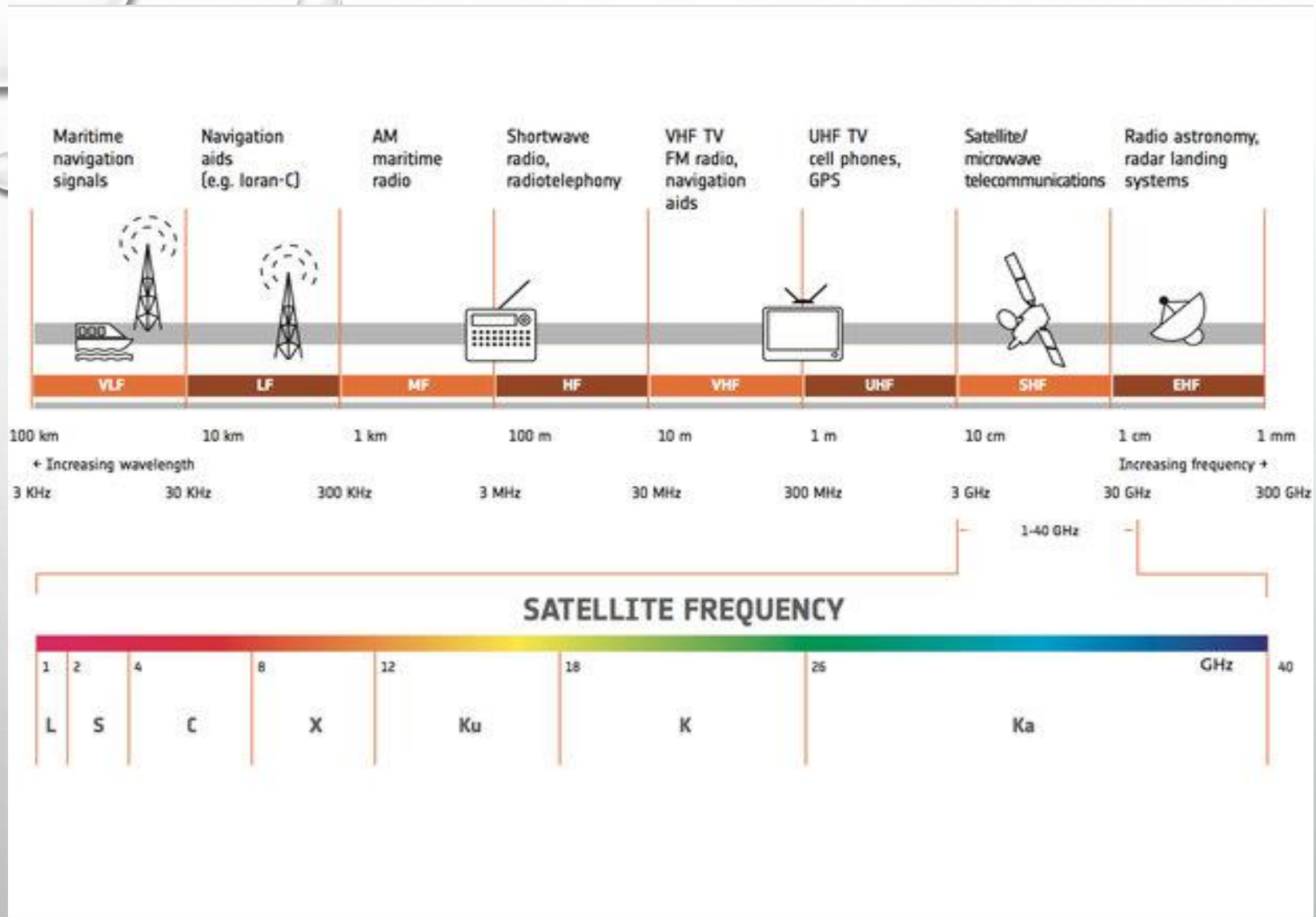


Figure: Message switching.

Bandwidth

Definition of bandwidth

- ₁ :a range within a band of wavelengths, frequencies, or energies;
- especially :a range of radio frequencies which is occupied by a modulated carrier wave, which is assigned to a service, or over which a device can operate.
- ₂ :the capacity for data transfer of an electronic communications system graphics consume more bandwidth than text does; especially :the maximum data transfer rate of such a system. A bandwidth of 56 kilobits per second



Bandwidth

f	λ	Band	Description
30–300 Hz	10^4 – 10^3 km	ELF	Extremely low frequency
300–3000 Hz	10^3 – 10^2 km	VF	Voice frequency
3–30 kHz	100–10 km	VLF	Very low frequency
30–300 kHz	10–1 km	LF	Low frequency
0.3–3 MHz	1–0.1 km	MF	Medium frequency
3–30 MHz	100–10 m	HF	High frequency
30–300 MHz	10–1 m	VHF	Very high frequency
300–3000 MHz	100–10 cm	UHF	Ultra-high frequency
3–30 GHz	10–1 cm	SHF	Superhigh frequency
30–300 GHz	10–1 mm	EHF	Extremely high frequency (millimeter waves)

